

VECTOR Reading Guide — Saracco + Bomiriya for the ASSIGN Mechanism

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Purpose: Distill the two papers most immediately relevant to replacing a bespoke ASSIGN model with a literature-grounded bipartite formation mechanism.

***Reading lens:** We are not asking either paper to solve the entire model. We are asking what structural or behavioral component each paper gives us, what it does not give us, and how the two might be combined with Quayle.*

Executive Synthesis

The two papers solve different pieces of our problem.

Saracco et al. (2015) gives us the **bipartite configuration-model skeleton**: preserve specified degree information on the two node layers while otherwise randomizing the bipartite network. This is directly relevant to a system in which individuals have one affiliation and teams have finite roster sizes.

Bomiriya et al. (2023) addresses the **homophily problem in bipartite networks directly**, without collapsing the bipartite graph to a one-mode projection. Its central message is especially important for us: homophily can be modeled while preserving the two-mode structure, rather than projecting first and losing information.

Quayle et al. (2006) supplies the behavioral idea we already like: similarity is an input to a stochastic attachment rule, while realized assortativity is an outcome.

Saracco: bipartite structural constraints + Quayle: similarity-weighted choice + Bomiriya: precedent for bipartite homophily

The simplest candidate ASSIGN rule remains:

$$P(i \rightarrow j) = \frac{R_j e^{-\rho d_{ij}}}{\sum_l R_l e^{-\rho d_{il}}}$$

where R_j is remaining capacity and d_{ij} is individual-to-team-centroid distance.

Important: That exact sequential rule is our proposed synthesis. It is **not claimed here to be an equation from either Saracco or Bomiriya**.

Paper 1 — Saracco et al. (2015)

Citation

Saracco, F., Di Clemente, R., Gabrielli, A., & Squartini, T. (2015). *Randomizing bipartite networks: the case of the World Trade Web*. Scientific Reports, 5, 10595. <https://doi.org/10.1038/srep10595>

The Problem They Are Solving

Saracco et al. are primarily concerned with **null models for bipartite networks**: what portion of the observed bipartite structure should be preserved while everything else is randomized?

For ASSIGN, the structural facts include one affiliation per individual at a snapshot and finite team sizes. Those constraints should not be confused with behavioral attraction.

Bipartite Representation

Layer 1: individuals Layer 2: teams/pools

An edge means individual i belongs to team j . This is more faithful to our formation problem than beginning with an individual-to-individual projection.

Configuration-Model Logic

For our baseline snapshot:

$$k_i = 1$$

for each individual, while a filled team has:

$$k_j = C_j.$$

For baseline NCAA men’s basketball:

$$C_j = C = 15.$$

The conceptual payoff is:

Roster size belongs in the structural constraint, not inside the homophily parameter.

Hard vs. Soft Constraint Warning

This distinction matters. Saracco's maximum-entropy BiCM constrains degree information at the **ensemble expectation** level. Our roster cap is operationally a **hard constraint**:

$$n_j \leq C_j.$$

If every roster position is filled:

$$n_j = C_j.$$

Therefore we can borrow the **configuration-model principle** without assuming that Saracco's particular soft-constraint sampling implementation is our final algorithm.

What Saracco Gives Us

- A defensible bipartite foundation.
- Degree information as the structural baseline.
- A clean null-model interpretation.
- Separation of degree/capacity effects from higher-order organization.
- A reason to preserve bipartite structure when it contains substantive information.

At $\rho = 0$, our natural analogue is a capacity-constrained random assignment baseline.

What Saracco Does NOT Give Us

Saracco does not directly supply:

- team-centroid similarity;
- Quayle's exponential homophily kernel;
- yearly roster turnover or tenure;
- evolving individual performance;
- lagged endogenous centroids;
- our exact sequential assignment rule;
- a single ρ controlling individual–team attraction.

Use Saracco as the structural parent, not as the complete ASSIGN model.

What to Mark While Reading

Look for passages on bipartite null models, degree sequences on both layers, maximum entropy, canonical/soft constraints, edge probabilities, independence of links, projection, and what degree constraints explain or fail to explain.

Reading question: Can we preserve the configuration-model structural logic while replacing unbiased edge formation with a similarity-biased matching process?

Paper 2 — Bomiriya et al. (2023)

Citation

Bomiriya, R. P., Kuvelkar, A. R., Hunter, D. R., & Triebel, S. (2023). *Modeling Homophily in Exponential-Family Random Graph Models for Bipartite Networks*. arXiv:2312.05673. <https://arxiv.org/abs/2312.05673>

The Problem They Are Solving

Bomiriya et al. ask how homophily should be modeled when ties connect **two different classes of nodes**. They explicitly address the common practice of collapsing a bipartite network into a one-mode projection, which can discard meaningful two-mode information.

That supports working with the actual affiliation process rather than forcing a one-mode model to represent it.

Why This Is Relevant

Our cross-mode comparison is:

individual performance $\longleftrightarrow$ team characteristic.

With a lagged endogenous centroid:

$$\mu_{j,t-1},$$

our dyadic distance is:

$$d_{ij,t} = |A_i^* - \mu_{j,t-1}|.$$

Bomiriya provides literature precedent for studying **homophily directly in the bipartite graph**.

ERGM Logic in One Sentence

Schematically, an exponential-family random graph model assigns probability to an entire network:

$$P(Y = y) \propto \exp\{\theta^\top g(y)\}.$$

Y is the random network, y a possible realization, $g(y)$ selected network statistics, and θ their parameters.

The lesson for us is not necessarily to fit a full ERGM. It is:

Bipartite structure does not prevent us from defining and modeling homophily.

Why Projection Is Not Necessary

Our causal question is:

Who affiliates with which team?

not merely which individuals become co-members after affiliation. Projection may remain useful as an outcome or diagnostic, but it need not be the formation mechanism.

What Bomiriya Gives Us

- Direct precedent for bipartite homophily.
- A warning against solving the problem by projection alone.
- A vocabulary for cross-mode similarity.
- Evidence that homophily can coexist with formal bipartite network modeling.

What Bomiriya Does NOT Automatically Give Us

It does not automatically provide:

- hard roster capacity;
- Quayle’s exponential distance kernel;
- lagged centroids;
- tenure turnover;
- sequential vacancy filling;
- our desired $\rho = 0$ random-assignment baseline;
- the exact weighted-stub algorithm.

Use Bomiriya as conceptual and methodological precedent, not necessarily as the ASSIGN engine.

What to Mark While Reading

Look for why projection is inadequate, definitions of bipartite homophily, how attributes on the two layers enter the model, cross-mode similarity statistics, degree structure, parameter interpretation, and computational complexity.

Reading question: Can we borrow their conceptual treatment of bipartite homophily without inheriting machinery we do not need?

Putting Saracco + Bomiriya + Quayle Together

Source	Piece we want
Saracco et al.	Bipartite configuration logic; degree/capacity as structural constraints
Quayle et al.	Similarity-dependent stochastic attachment; preference $\neq$ realized assortativity
Bomiriya et al.	Direct bipartite homophily; projection is not required

A defensible description is:

ASSIGN is a constrained bipartite configuration-style assignment process augmented with a Quayle-inspired similarity bias.

Candidate Minimal Mechanism

Let $R_{j,t}^{(s)}$ be remaining openings on team j at assignment step s and define:

$$d_{ij,t} = |A_i^* - \mu_{j,t-1}|.$$

Then:

$$P(i \rightarrow j \mid t, s) = \frac{R_{j,t}^{(s)} e^{-\rho d_{ij,t}}}{\sum_l R_{l,t}^{(s)} e^{-\rho d_{il,t}}}.$$

After assignment:

$$R_{j,t}^{(s+1)} = R_{j,t}^{(s)} - 1.$$

When $R_{j,t}^{(s)} = 0$, the team has no remaining opening and receives zero probability.

Why Remaining Capacity Matters

At $\rho = 0$:

$$P(i \rightarrow j) = \frac{R_j}{\sum_l R_l}.$$

This is the configuration-style random matching baseline: an individual stub is matched to the remaining team stubs. Similarity then biases that baseline:

$$\text{available team stubs} \times \text{similarity attraction}.$$

Conceptual Payoff

$$C_j = \text{structural capacity}$$

$$\mu_{j,t-1} = \text{lagged endogenous team state}$$

$$d_{ij,t} = \text{individual-team difference}$$

$$\rho = \text{strength of similarity preference}$$

$$\text{realized assortment} = \text{outcome of constrained matching}$$

Therefore:

$$\rho \neq \text{realized assortativity}.$$

Capacity can prevent strong homophilic preference from producing equally strong realized sorting. That may be an important property rather than a defect.

Cautions Before Locking the Model

1. **Saracco’s BiCM is not identical to hard roster filling.** Be precise about whether the final implementation is a BiCM, hard bipartite configuration model, or configuration-style sequential matching algorithm.
2. **The weighted-stub equation is presently our synthesis.** The factor $R_j e^{-\rho d_{ij}}$ is parsimonious, but we should verify whether an established model already uses the exact or equivalent form.
3. **Do not import Quayle’s analytic $\alpha \leftrightarrow r$ mapping.** Our bipartite capacity constraints change that relationship. Initially establish $\rho \rightarrow$ realized sorting through simulation.
4. **Projection can remain a diagnostic.** It need not be the formation mechanism.

Suggested Reading Order

Saracco first — structure. Ask what exactly the bipartite configuration model preserves and whether constraints are hard or expected.
Bomiriya second — behavior. Ask how homophily is defined when the node classes differ, and which ideas can be borrowed without a full ERGM.
Quayle third — mechanism. Re-read Equation (4) asking which component is structural, which is similarity preference, and what should replace $f(k_j)$ for a finite-capacity team.

Five-Year Memory

Saracco gives us the container. Quayle gives us the attraction. Bomiriya tells us it is legitimate to study that attraction without destroying the bipartite network.

Protect this question throughout development:

How much realized sorting emerges from a given similarity preference once finite team capacities constrain what can actually be realized?

Sources

Saracco, F., Di Clemente, R., Gabrielli, A., & Squartini, T. (2015). *Randomizing bipartite networks: the case of the World Trade Web*. Scientific Reports, 5, 10595. <https://doi.org/10.1038/srep10595>

Bomiriyā, R. P., Kuvelkar, A. R., Hunter, D. R., & Triebel, S. (2023). *Modeling Homophily in Exponential-Family Random Graph Models for Bipartite Networks*. arXiv:2312.05673. <https://arxiv.org/abs/2312.05673>

Quayle, A. P., Siddiqui, A. S., & Jones, S. J. M. (2006). *Modeling network growth with assortative mixing*. European Physical Journal B, 50, 617–630.